Problem Chosen	2024 MCM/ICM Summary Sheet	Team Control Number
F		2416809

Wide future for wildlife

Summary

Did you hear that elephants were weeping on the Masai Mara prairie? Did you see that the poachers' bullets pierced Verde's rhinoceros's thick skin? **No business, no harm**. Now, the total amount of illegal wildlife trade in the world has reached about 26.5 billion US dollars per year and is considered to be the fourth largest of all global illegal trades. How many lives does this cost? How many scars have been left on our earth?

To crack down on IWT, our team designed **a data-driven project** for target clients to predict the location of crime through DNA tracking and machine learning, and equipped with a data sharing platform and UAV system to assist law enforcement, to drive win-win cooperation among regions.

For task 1, we developed a **Client Anchoring Model (CAM)**, which calculates the weights of three primary indicators and nine secondary indicators based on the **AHP-EWM** model, and uses **GE Cube** to visualize the results for preliminary judgment. Finally, we calculated the weighted comprehensive client index **(CCI)** and concluded that our best customer is a developing country with IWT.

For task 2, we first targeted our clients in eastern and southern African countries, designed our projects, and proposed budgets and necessary conditions. At the same time, we use the **Gantt chart** to reflect the process of a **5-year project** in stages. To convince our clients, we used **BP neural network** and InterVention Model (IVM) to predict that the overall number of poaching incidents in Africa will decrease by about 397 after the implementation of the project.

For task 3, IWT is considered in the global market system. We suggest cooperating with developed countries to adjust the demand side, carrying out financial financing on a global scale, and providing additional resources and power for the project through multimedia marketing.

For task 4, based on task 3, the demand for IWT decreases over time under the global market system. Based on the principles of game theory and cost-benefit analysis, we created the **Time-ordered Anti Poaching Game Model (TAPGM)**. The results show that the expectation of poachers will drop to -22.6 year by year, and the number of poachers will also decrease year by year. At the same time, the project will also create a total utility of 488 million dollars for the government, which further convinces clients that our project is appropriate for them.

For Task 5, We optimize the BP neural network prediction using an **Improved Genetic Algorithm (IGA)** and stably calculate the likelihood of the project achieving the expected goals. The results show that when the investment in funds increases by 20%, we expect an 87.9% chance of achieving the expected goal.

In summary, our comprehensive method promises a sustainable future for wildlife conservation.

Keywords: IWT；data-driven; AHP-EWM;GE Cube;Game Theory; IGA optimized BP network

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1 Introduction

1.1 Problem Background

With the progress of society, human beings have deepened their exploration of our Earth, but many phenomena are contrary to the advancement of society. One such phenomenon is the illegal wildlife trade (IWT), which pushes many animals to the brink of extinction. It also disrupts the **food chain**, **global biodiversity**, and the **entire ecosystem**…

In addition to the damage to nature and ecology, IWT has a significant impact on human society. For instance, IWT is utilized for **corrupt purposes** such as money laundering and bribery, endangering national governance. The poaching and illegal trading of wild animals also raise the risk of **species invasion**. Moreover, **infectious diseases** may emerge, greatly affecting public health and the normal operation of human activities.

Figure1: A cute little baby monkey was trapped in a cage due to IWT

The damage caused by IWT is irreversible. If it is allowed to develop without control now, countries and the world will incur greater expenses in the future for repairs. Therefore, to reduce the emergence of IWT and improve the well-being of all the species in the world, we have designed the following project to solve this problem and provide relevant clients with pertinent policy advice, strong data support, and analysis.

1.2 Restatement of the Problem

Considering the background and the scientific nature of the project's implementation, we summarize the task as **3D2A**:

- Task 1: Define our target client.
- Task 2: Design a project for the client and convince them.
- Task 3: Discuss additional resources and capabilities the client will need.
- Task 4: Assess the impact after the project implementation.
- Task 5: Analyzing the possibility of achieving expectations and considering other influencing factors

1.3 Our Work

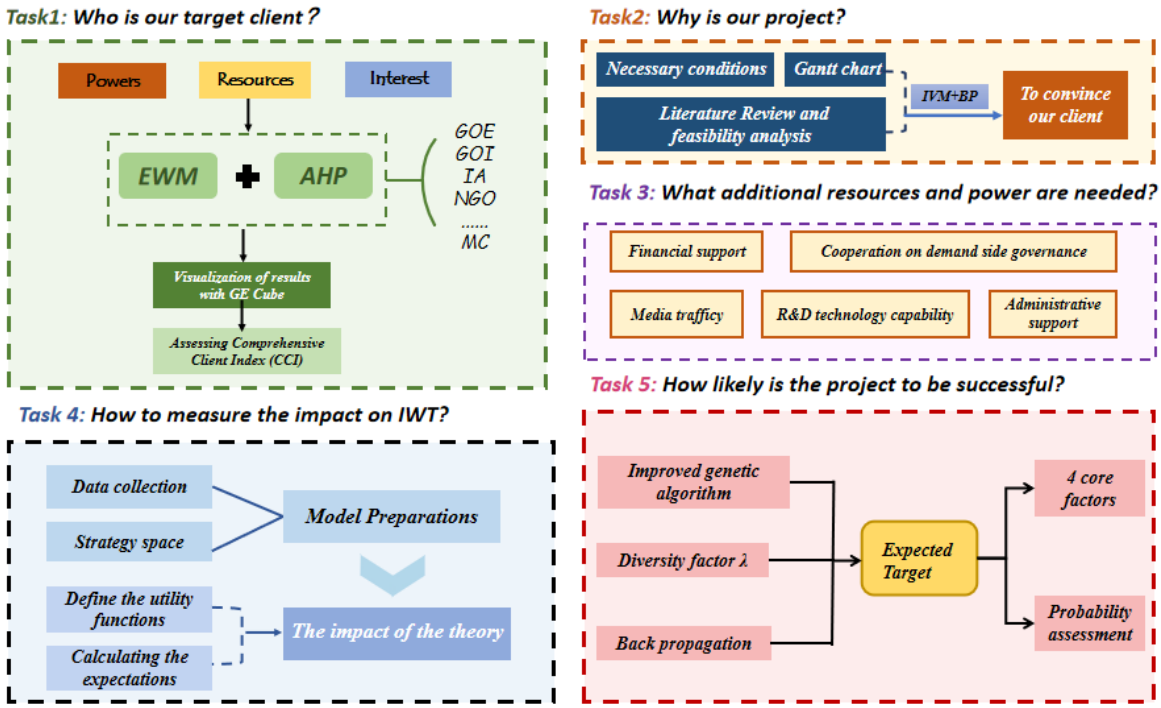

Figure 3: The structure of Our Work

2 Assumptions and Justifications

2.1 Assumption for overall

- We assume that the average level of historical data has credibility

  The specific implementation of the project requires further examination of the national conditions and practical difficulties of each country. Therefore, we have selected historical average data for estimation, proposed a budget, and estimated future benefits.

2.2 Assumption for task 2

- The impact of intervention measures on illegal trade measures is non-linear.
- During the project implementation period, there will be no significant external changes (such as laws, regulations, etc.).

2.3 Assumption for task 3

- Assuming that the IWT market is a perfectly competitive market.

  Because there are many buyers and poachers in the IWT market, neither side can completely control prices through monopoly, so it can be simplified as a perfectly competitive market.

2.4 Assumption for task 4

- We assume that both the government and poachers are rational economic men.

  Both the government and poachers make decisions based on strategies to maximize their own utility.

- We don't consider the strategy of the public reports.

  The behavior of the public will promote the implementation of the project, but it will not directly affect the decisions of the government and poachers.
- We don't consider the model as a repeated game.

  The formulation of policies has continuity. Policies won't change in a short period. So, we can assume that the government will only make one decision within five years.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description	Unit
U_P	The total utility of poachers	Million dollars per year
U_G	The total utility of the government	
U_{G-0}	The utility of the government before the project implemented	
U_{G-t}	The total utility of the government in the *t-th* year	
U_{P-t}	The total utility of poachers in the *t-th* year	
U_{pos}	The positive externalities of reducing IWT	
U_{neg}	The negative externalities of IWT	
R_p	The profit of poachers	
p_p	The probability of successful poaching by poachers	/
p_c	The probability of successful capture by government (1-p_p)	
p_I	The probability of government's effective enforcement	
C_p	The cost of poachers when they been captured	Million dollars per year
C_I	The cost caused by project content	
C_a	The administrative costs of implementing projects	
C_h	The manpower cost of implementing the project	
C_u	The cost of building and maintaining the drone system	
C_e	The cost of building and maintaining a data platform	
C_r	The cost of rewarding the public for reporting	
C_m	The cost of maintaining project hardware and platform	
C_t	The technical cost of conducting crime prediction	
E_P	The expectations of poachers	
D	The demand of illegal wildlife products	Million dollars
P	The average price of illegal wildlife products	
t	The *t-th* year when the project is implemented	year

4 Task 1: Who is our target client?

Although IWT is a global problem, not all governments and social organizations can effectively and willingly implement strategies to reduce the occurrence of IWT.

In order to ensure the effectiveness and continuity of the 5-year project, we set up a **Client Anchoring Model (CAM)** based on 3 categories (primary indicators) and 9 aspects (secondary indicators) using **AHP-EWM** to calculate the weight of indicators at each level.

We also constructed a **GE Cube** based on the GE matrix to visualize our evaluation results and obtain a preliminary estimate.

Finally, we added weight accuracy to calculate the **Comprehensive Client Index (CCI)** to quantify our comprehensive scores for various indicators.

4.1 Indicators of Client Anchoring Model (CAM)

Client assessments can be divided into 3 categories based on the background of the problem and the reality of the situation.

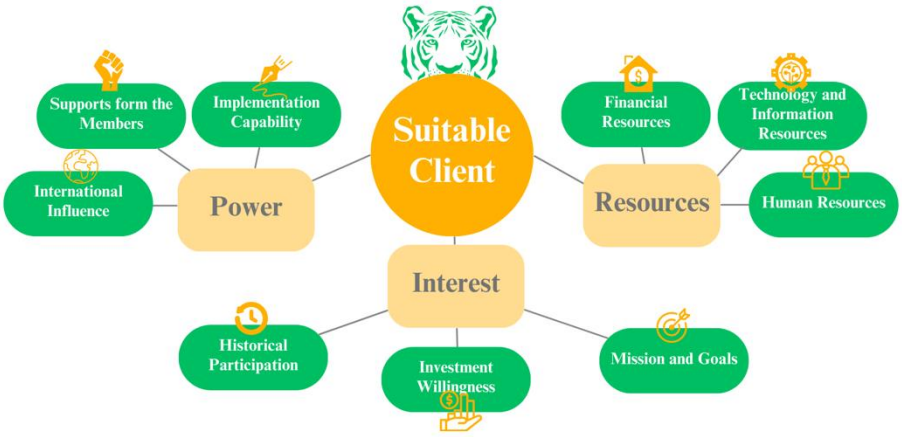

Figure 4: The structure of Client Anchoring Model

- **Power:**

Whether it is a country or a social organization, it must have sufficient execution efficiency and effectiveness to implement policies and projects. Power can also be divided into internal and external. **Internal power** is reflected not only in the **top-down** instruction transmission and execution effectiveness, but also in the **bottom-up** support from members within the unit collective for combating IWT. **External power** refers to the international influence of the country's government or organization.

  - *Implementation Capability (IC)*
  - *Supports from the Members (SM)*
  - *International Influence (II)*

- **Resources:**

The resources that the client can provide are the material guarantee for the project. **Financial resources** are the most critical indicator, which provides the prerequisite for the implementation of all projects. With advanced **technology and information resources**, the client can capture the dynamics of wildlife and trading activities faster and make decisions efficiently. It also requires the client to have enough **human resources** to deploy and ensure more people can devote themselves to wildlife protection.

  - *Financial Resources (FS)*
  - *Technology and Information Resources (TIR)*
  - *Human Resources (HR)*

- **Interest:**

If the client has power and resources but does not have enough willingness to cooperate with the implementation of the project, everything is empty talk. Since client's attention and

emphasis on IWT will not change significantly within a period (administration is stable), we can obtain the score for these indicators based on reasonable speculation.

- *Historical Participation (HP)*
- *Investment Willingness (IW)*
- *Mission and goals (MG)*

4.2 The Establishing of CAM

According to the question, we have listed seven target clients as candidates, namely: government (developed countries) (GOE), government (developing countries) (GOI), independent agency (IA), international organization (IO), non-governmental organization (NGO), multinational corporation (MC), and media unit (MU).

The data we use to calculate weights comes from multiple databases [1]. We select data from 2019-2023, so that the validity of the data is also 5 years. We also referred to some relevant literature and used to score based on experts' opinions.

4.2.1 Weighting primary indicators by AHP

First, we analyzed relevant literature and expert opinions, scored the primary indicators objectively and rationally, and formulate a comparison matrix.

Next, we conduct a consistency test on the constructed matrix and obtain CR. The result of CR=0.0176<0.1, so the consistency of the comparison matrix is acceptable.

In order to ensure the scientific of the weights, we used the arithmetic mean method, the geometric mean method and the eigenvalue method to calculate three sets of weights respectively. **Averaging the three sets of weights** is our result (shown in *Table 2*).

4.2.2 Weighting secondary indicators by EWM

Since our data are already normalized and there are no negative numbers, we can perform calculations directly without standardization. From this, we established a normalized matrix Z_{n*m}(n represents the object to be evaluated, m represents the evaluation index).

Next, we calculate the proportion of the *i-th* sample under the *j-th* indicator. Regarding it as the probability used in the relative entropy calculation. We calculate the probability matrix P_{m*n}. The calculation formula for each element P_{ij} of P_{m*n} is as follows:

$$P_{ij} = \frac{\tilde{Z}_{ij}}{\sum_{i=1}^{n} \tilde{Z}_{ij}} \tag{1}$$

Calculate the information entropy of each index:

$$e_j = -\frac{1}{\ln n} \sum_{i=1}^{n} p_{ij} \ln(p_{ij}), (j = 1, 2, \dots, m) \tag{2}$$

Calculate information utility value:

$$d_j = 1 - e_j \tag{3}$$

Through normalization, we obtain the entropy weight of each secondary indicator, and the calculation formula is as follows:

$$W_j = \frac{d_j}{\sum_{j=1}^{m} d_j}, (j = 1, 2, \dots, m) \tag{4}$$

The final weight we calculated by combining AHP and EWM is as follows:

Table 2: Weight of Index

	Indicators(I)	Weights	Indicators (II)	Weights
Suitable Client	Power	0.3198	Implementation Capability (IC)	0.6243
			Supports from the members (SM)	0.2812
			International Influence (II)	0.0945
	Resources	0.1222	Financial Resources (FR)	0.5815
			Technology and Information Resources (TIR)	0.3090
			Human Resources (HR)	0.1095
	Interest	0.5580	Historical Participation (HP)	0.1368
			Investment Willingness (IW)	0.6242
			Mission and Goals (MG)	0.2390

4.2.3 Visualization of results with GE Cube

Because our evaluation **indicators are three-dimensional**, the traditional GE matrix showing two-dimensional variables cannot meet our visualization requirements. On this basis, we constructed a GE Cube. It allows us to have a preliminary estimate of the scores of each client (shown in *Figure 5*).

Figure 5: GE Cube in Client Anchoring

According to charts, we can know that the governments of developed countries have the highest scores in power and resources, but insufficient interest. Although the interest of international organizations and non-governmental organizations is high, their power and resources are difficult to support the implementation of mandatory measures. What can be found is that the **governments of developing countries** have relatively high comprehensive scores in various indicators. They should be more suitable for our project.

4.3 Assessing Comprehensive Client Index (CCI)

Our basic conclusion drawn from the visualization results of the GE Cube is to select governments from developing countries as our selected clients. However, the data visualization is **not rigorous enough**, so we further constructed a comprehensive client index CCI to help us draw the conclusion. The calculation principle is as follows:

$$W_j = \sum_{i=1}^{m} w_i w_j \tag{5}$$

$$CCI_i = \sum_{i=1}^{m}\sum_{j=1}^{n} W_j x_{ij} \tag{6}$$

In the above two equations, W_j is the weight vector of the secondary indicator relative to the global, w_i is the weight vector of the primary indicator relative to the global, w_j is the weight vector of the secondary indicators relative to the primary indicators, m is the number of primary indicators, n is the number of all secondary indicators, CCI_i is the comprehensive client index of the *i-th* client to be selected, x_{ij} is the score of the *i-th* client in the *j-th* indicator.

Through the above calculation process, we can obtain the scores of seven target clients as follows:

CCI= [88.1766, 91.2133, 86.4632, 90.4671, 89.2037, 86.1784, 82.7237]

Figure 6: The CCI of different clients

According to CCI, our best clients should be **developing country governments** with a history of governing IWT. We found that countries in Southeast Asia and eastern and southern Africa were more significant [2].

To make our project more targeted, we have selected relevant departments and organizations under the governments of some countries in **eastern and southern Africa** as clients (such as Kenya, South Africa, Tanzania, Uganda, etc.) to carry out the project.

5 Task 2: Why is our project?

5.1 What is our project?

5.1.1 The content of our project

- **Using deep learning algorithms and DNA tracking to predict crime locations**

By using the public application programming interface (API) to access Internet information and the **deep learning model**, the government can detect suspicious wildlife transactions from a large amount of data, analyze the historical data of IWT, predict and effectively crack down on criminal gangs.

DNA testing can be carried out on captured illegal wildlife products to trace the source of DNA and find hunting grounds for wildlife. Then, through **data analysis** and other predictive models, advance prevention of criminal behavior in the predicted area can be realized.

- **Divide all areas inhabited by wildlife into designated law enforcement personnel and armed with drone inspections**

To enhance regulatory efficiency, we will equip grassroots law enforcement personnel with drone equipment. Drones not only **monitor illegal activities**, but also **archive data** on large terrestrial wildlife settlements in their respective areas. The data mainly includes the number of organisms and migration routes. Once an abnormal vacancy or death occurs, an emergency search is required.

- **Building a digital sharing platform and a public reporting mechanism**

Build a digital platform that can store survival information of wild animals. A **shared platform** can allow more people to participate in the protection of African wildlife, such as setting up **virtual animal adoption** and allowing third parties to participate in monitoring activities.

By **providing rewards for reporting** to the public, we encourage them to participate in animal protection.

- **Establishing the special cooperation between governments**

From the administrative distribution in Africa, there are lots of countries in the east and south of Africa. There is a blind spot in the jurisdiction of IWT in the national border area. Therefore, we need to strengthen cooperation in **customs and border enforcement**.

At the same time, the large number of African countries has also led to inconsistent standards and means of punishing IWT, and governments also need to unify the **legal system** for governing IWT.

5.1.2 Budgets and necessary conditions (for a single country)

Table 3: Budgets and necessary conditions

Conditions	Entries	Amount
Fund	The administrative costs of implementing projects	10 million
	The manpower cost of implementing the project	20 million
	The cost of building and maintaining the drone system	80 million
	The cost of building and maintaining a data platform	60 million
	The cost of rewarding the public for reporting	30 million
	The cost of maintaining project hardware and platform	50 million
	The technical cost of conducting crime prediction	30 million
Infrastructure	Unmanned area network coverage	/
	Public education	
	Allocation of grassroots law enforcement personnel	
System	Data analysis technology system	2
	Data platform reporting system	1
	Grassroots law enforcement three-level management system	1
	Drone Network System	1

5.1.3 Gantt chart of project

Here, we use a Gantt chart to visualize the project timeline. By using this visual guide, it is possible to gain a faster understanding of the project's progress and phased results。

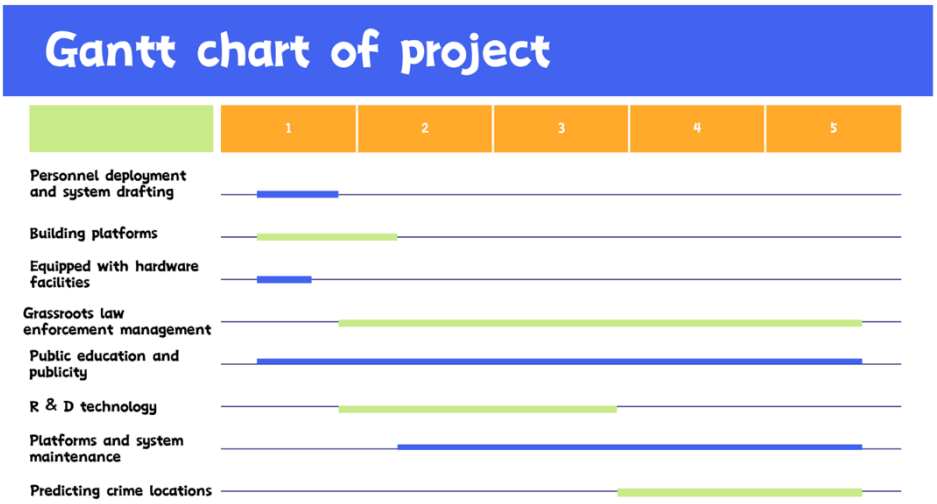

Figure 7: Gantt chart of project

5.2 Literature Review and feasibility analysis

To demonstrate the feasibility and scientific of the project, we collected relevant literature and academic research results to provide material support for our project.

- **Using deep learning algorithms and DNA tracking to predict crime locations**

I In the article "*International Wildlife Traffic: A perspective on the challenges and potential forensics genetics solutions*" published by *Utpal Smart* [3], the important role of DNA detection in identifying IWT behavior is proposed. Based on this technology, we can track and trace the origin of organisms based on DNA information and use DNA databases to infer high crime incidence areas and predict future crimes.

In addition, in the article "*Use of Machine Learning to Detect Wildlife Product Promotion and Sales on Twitter*" [4], the author uses (API) to access messages. By combining unsupervised machine learning topic models with keyword filtering and manual annotation, suspicious IWT can be detected. This also indicates that the government can indeed detect IWT from a large amount of data, analyze historical data of illegal poaching and trading.

- **Divide all areas inhabited by wildlife into designated law enforcement personnel and armed with drone inspections**

In the article: "*Are unmanned aerial systems (UASs) the future of wildlife monitoring? A review of admissions and challenges*" [5], the author mentioned that compared with traditional satellite, unmanned aerial vehicle remote sensing has advantages. *R. Raffik et al.* [6] also proposed that drones can also detect strange behaviors in human activities. **Deep learning** algorithms can be used to identify certain animals, track their actions, and classify their behavior. By utilizing this knowledge, one can better understand the ecology of wildlife populations, and it can also be used to develop more successful conservation plans.

- **Building a digital sharing platform and a public reporting mechanism**

According to the official website of WWF [7], there is a column on virtual adoption of wildlife, and the sales are high. Therefore, we can infer that the development of virtual adoption

has a market driven by the traffic of African wildlife.

In the *World Wildlife Crime Report* released by *the United Nations* [8], it is mentioned that when authorities proactively record these details, a large amount of information about the identity and methods of criminals can be collected. The government can use its power to record in detail all poachers and their illegal activities on this platform and provide generous rewards and subsidies to whistleblowers who implement the situation based on the authenticity and effectiveness of the information.

5.3 Project suitability analysis

According to "These Countries Are At The Center of The Illegal Wildlife Trade" published in the Proceedings of the National Academy of Sciences, we can know that numerous countries with IWT is mainly located in East Africa and Southeast Asia, as shown below:

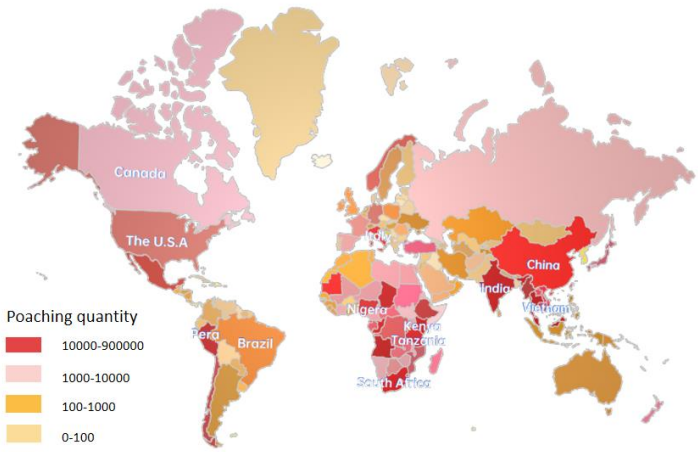

Figure 8: The number of IWT between countries

In order to illustrate the suitability of our project, we selected some countries and regions in **southeastern Africa** for analysis.

The specific contents are as follows:

First, we found four types of data from 2004 to 2023 in the working documents of CITES' Eighteenth meeting of the Conference of the Parties, the World Wildlife Crime Report issued by the United Nations, and other international organizations and government websites: poaching incidents in Africa quantity, the number of illegal wildlife products seized by UNODC in Africa, the number of rhino deaths (due to poaching) in southeast Africa and the inflation-adjusted average price of live white rhinos.

Next, use the four data as the input of the BP neural network, select ReLU as the activation function, set the number of hidden layers to 3, and calculate the error through MSE for back propagation:

$$MSE = \frac{1}{2N}\sum_{i=1}^{N}(target_i - output_i)^2 \tag{7}$$

In the above formula, MSE is Mean Squared Error, N represents the number of samples, $target_i$ is the target output of the *i-th* sample, $output_i$ is the actual output of the *i-th* sample.

Then in backpropagation, the gradient of the error with respect to the weight is calculated:

$$\frac{\partial E}{\partial w_{ij}} = -(target_i - output_i) \cdot f'(net_j) \cdot y_i \tag{8}$$

In the above formula, $\frac{\partial E}{\partial w_{ij}}$ represents the gradient of the error for the weight of connection i to j, $f'(net_j)$is the derivative of the activation function, y_i is the input of connection i.

Finally, the weights are updated via gradient descent:

$$w_{ij} = w_{ij} - \eta \frac{\partial E}{\partial w_{ij}} \tag{9}$$

In the above formula, η is the learning rate, which controls the step size of weight update.

After inputting the data into the neural network for fitting and obtaining the prediction results, in order to simulate the intervention effect of the project, we constructed an **"Inter-Vention Model" (IVM)** containing "intervening variables" and "intervention relationships". We consider the following impacts after the implementation of the project: increased law enforcement by relevant personnel in the area; improved monitoring capabilities and accuracy on the number of wild animals and illegal wildlife poaching; increased satisfaction of community needs and increased sense of participation among community residents, among other impacts.

Through IVM, we can calculate the values of each of our projects after the project is implemented. The specific relationship is as follows:

$$P_t' = P_t + \gamma_E \times E_t + \gamma_M \times M_t + \gamma_C \times C_t + \varepsilon_t' \tag{10}$$

In the above formula, P_t' is the value of each indicator after taking into account the intervention effect, the intervention variables E_t、M_t and C_t are the quantitative values of law enforcement intensity, monitoring capacity and community participation in year t respectively, γ_E、γ_M and γ_C are these The coefficient of the impact of intervening variables on the number of poaching hours, ε_t' is the error term after taking into account the intervention effect.

Our establishment of IVM is based on the following considerations:

① The parameters of the linear model we use are relatively simple, which can help non-professional policymakers understand our model and impact; ② Previous research cases from various countries around the world mostly use linear models to evaluate the effects of policy intervention, which provides us with the opportunity to introduce IVM precedent.

Based on the specific national conditions of southeastern African countries, we set E_t to 1.5 (expected to increase law enforcement intensity by 50%); set M_t to 2.0 (expected to increase monitoring capabilities by 2 times); set C_t to 1.3 (expected to increase community participation by 30%). After bringing P_t into the above formula, the comparison of various indicators after the implementation of the project is calculated, as shown below:

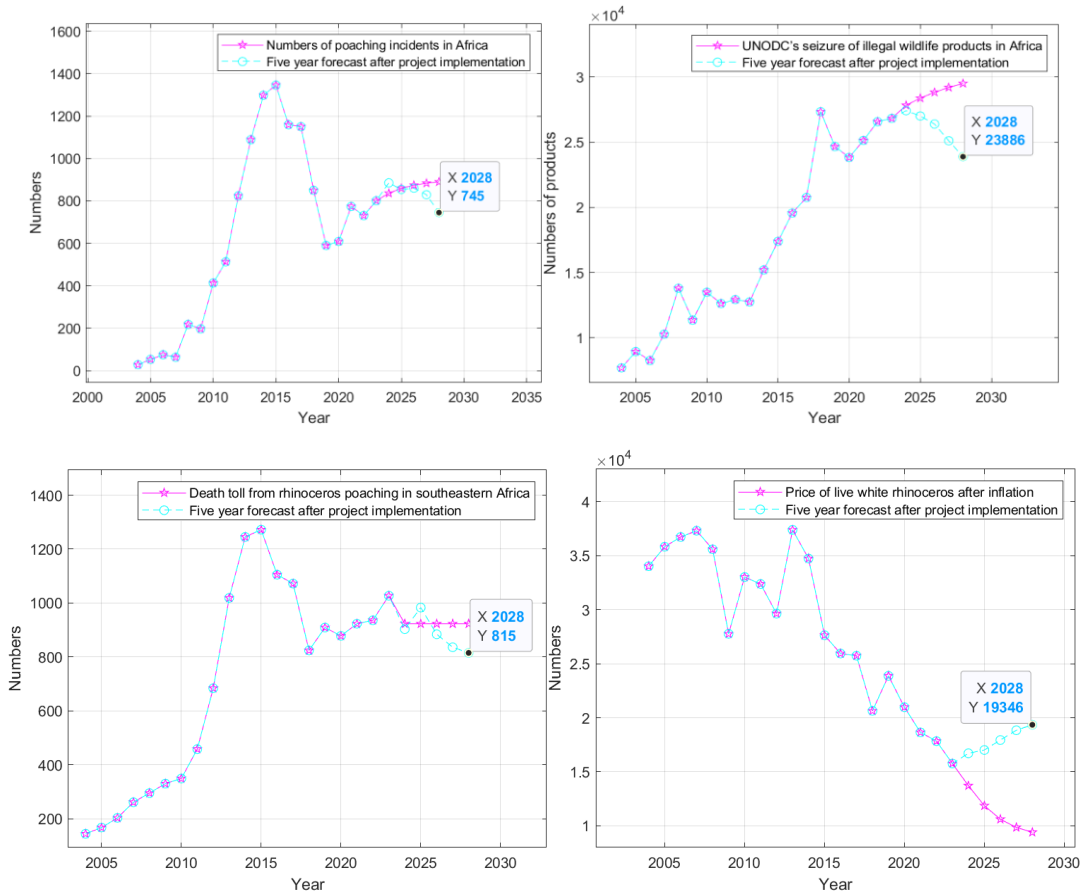

Figure 9: The prediction of these four indicators

Preliminary conclusions can be drawn from analyzing the prediction results of the above four indicators:

After the fifth year of project implementation, the number of poaching incidents in Africa will decrease by **397** compared to expected, and the illegal wildlife and plant products seized by UNODC will decrease by **3951** compared to before. The number of rhinoceroses killed due to poaching in southeastern Africa will decrease by **786** compared to when the project was not implemented. For the price of live white rhinoceros adjusted for inflation, the price per live white rhinoceros will be **$3463** higher than when the project was not implemented.

Based on the above prediction results, we can draw the following conclusion: if our project is not adopted, illegal wildlife poaching and trade in the southeastern region of Africa will continue to exist and become increasingly serious; If we adopt our project, although the effect may not be significant in the short term, as a relatively long-term 5-year plan, it will have significant effects in 4-5 years after the implementation of the project. Illegal wildlife crime and trade related behaviors will be significantly reduced, and with the recovery of species populations, the stability and biodiversity of the ecosystem will be more stable.

In summary, our project is very suitable for our clients and our clients should undertake our project.

6 Task 3: What additional resources and power are needed?

To make our suggestions more targeted, we have locked our clients in Kenya, South Africa,

Tanzania, and Uganda, which are the countries with the most severe IWT.

6.1 Financial support

Based on the collected historical economic data and prediction of the financial situation of the four governments by scholars [9], we can infer that these governments will implement expansionary fiscal policies in the long term (excluding South Africa).

We also searched the official websites of various governments for budget reports and fiscal expenditure structures [10] [11]. It was found that their expenditure on ecological protection, when converted into US dollars, can't fully cover the cost of promoting the project in the first year (but there is a certain chance of covering the cost of the next four years).

Therefore, we suggest that:

International organizations with social responsibility and non-governmental organizations (such as the United Nations and WWF) can provide financial and human assistance. The World Bank can also provide special loans to countries and provide long-term support by lowering interest rates.

6.2 Cooperation on demand side governance with developed countries

In the assumption, we assume that the IWT market is a perfectly competitive market, so the market can only reach equilibrium when supply equals demand.

Most illegal wildlife product transactions are cross-border. In order to address the prevalence of IWT, it is necessary not only for developing countries to make efforts in combating poaching, but also for global efforts to address the demand side.

We suggest establishing demand side governance cooperation with developed countries. Once demand decreases and supply exceeds demand, the prices of illegal wildlife products decrease, resulting in a decrease in the profits of poachers.

Specific forms of cooperation can be referred to：

- Production of substitutes (such as through artificial leather goods, etc.) [12]
- Strengthen domestic governance of corruption and promote financial transparency
- Agree on controversial wildlife products or impose higher consumption taxes
- Increase legal penalties for purchasing illegal wildlife products

6.3 R&D technology capability

Our project is data-driven, so it requires professional researchers to analyze data and conduct research. The country needs a certain level of research strength.

If there is no technology or talent in **DNA traceability**, **machine learning**, **drone management system**, etc., it is necessary to seek external assistance.

For example, exchanging research projects with some universities or collaborating with research institutions in the market to achieve project implementation.

6.4 Media traffic

The role of the media cannot be ignored!

Although our project is designed for the government, it also requires the support of the entire society. The media have a huge influence and audience, which can promote our projects to more places and arouse more people's resistance to wildlife products.

- In the era of fragmented information, we can use streaming media platforms (such as TikTok, Instagram, etc.) for short video promotion and live delivery. It can also bring more people to adopt our virtual wildlife and inject more active funds into the project.
- Some traditional media (such as BBC, National Geographic, etc.) also need to exert influence and bring more social attention to the fight against IWT through news, movies, or publications.

6.5 Administrative support

To ensure the efficient implementation of the project and reduce administrative and decision-making delays, we suggest setting up a **dedicated execution department** under the leadership of the government or providing smooth administrative channels.

7 Task 4: How to measure the impact on IWT?

The government fights against IWT by implementing projects. The government doesn't know what decisions the poachers will make, but in general (i.e., with a higher probability) the government will implement the project.

However, not 100% of the project is implemented perfectly, and there is a certain **probability of implementation leakage**. Thus, poachers can make decisions based on the expected efficiency of the government's implementation of the project, allowing them to maximize their expected behavioral choices.

To address this specific problem, we improve the traditional game model by treating this decision-making process between the government and the poacher as an incomplete mixed-strategy game process. Therefore, we design a **Time-ordered Anti-Poaching Game Model (TAPGM)** to compare the changes in poachers' benefits before and after the implementation of the project and to analyze the results five years after the implementation of the project.

7.1 Establishing the TAPGM

7.1.1 Strategy space

Table 4: The strategy space of the government and poachers

		Government	
		Implement (I)	Not Implement (NI)
Poachers	Poach (P)	(P, I)	(P, NI)
	Not Poach (NP)	(NP, I)	(NP, NI)

7.1.2 Define the utility functions

- **The utility function of the poacher:**

$$\begin{cases} U_P = R_p \cdot p_p - C_p \cdot p_c \\ R_p = D \cdot P \end{cases} \tag{11}$$

The total utility of poaching should be equal to the total benefits received by all poachers if poaching is successful, minus the paid if they are caught. The level of benefits depends on the demand in the IWT market.

- **Considering the impact of supply and demand in the IWT market based on the background of Task 3:**

In Task3, we discussed that cooperation with developed countries (e.g., by adopting behaviors such as educational campaigns, producing production of substitutes) leads to a decrease in the demand side of the IWT. Thus, *D* is a decreasing function over time.

Since we state in our assumptions that the IWT market is a perfectly competitive market, supply equals demand in order to reach equilibrium.

It is reasonable to assume that the benefits received by poachers over the five years of the policy are also an accelerating decreasing over time:

$$R_{p-t} = 600 - 10t^2 \tag{12}$$

Here, our initial values for the first year are modeled using the total amount of illegal hunting in Africa estimated by the United Nations for all years.

In summary, we can establish a similar link between the parameters, showing the change in total utility after project implementation in the case that time affects demand.

$$U_{P-t} = R_{p-t} \cdot p_p - C_{p-t} \cdot p_c \tag{13}$$

- The utility function of the government:
  - The first year:

$$\begin{cases} U_G = (R_P + U_{pos}) \cdot p_c - U_{neg} \cdot P_p - C_I \\ C_I = C_a + C_h + C_u + C_e + C_r + C_t \end{cases} \tag{14}$$

Due to **externalities**, combating the IWT as part of a complex system results in positive and negative externalities. Thus, the benefits to the government come from the value and additional welfare gained from catching poachers.

The costs to the government are the negative externalities from the small number of uncaught illegal poachers and the costs of implementing the project.

In the first year, the main reason is that the construction cost of the project is relatively high, soC_g, C_u, and C_e are all high.

  - The rest four years:

$$\begin{cases} U_G = (R_P + U_{pos}) \cdot p_c - U_{neg} \cdot P_p - C_I \\ C_I = C_a + C_h + C_m + C_r + C_t \end{cases} \tag{15}$$

However, starting from the second year, the cost of establishing drone systems and platforms will decrease. Moreover, the administrative and labor costs incurred by the government to promote the project will also be significantly reduced.

In the following four years, only lower administrative costs, maintenance costs, and quantitative reward costs need to be paid.

By the same token above, it is also possible to extrapolate the utility of ug under the time

utilityU_{G-0} and U_{G-t}.

7.1.3 Calculating the expectations before and after project implementation

Since the government has a predominant position, however, the government's policies may not always be fully implemented (i.e., there is still a small amount of corruption and bribery). Therefore, we weight the percentage of project that can be implemented effectively as P_I. Poachers can only make judgments based on their predictions for the probability (P_I) that the government will effectively implement the project. The expectation of poachers is as follows:

- If poaching:

$$E_P = U_I \cdot p_I - U_{NI} \cdot (1 - p_I) \tag{16}$$

- If not poaching:

$$E_P = 0 \tag{17}$$

At the beginning, the government has made decisions according to its own effectiveness, while poachers follow the principle of maximizing their expectations and choose the most appropriate strategy.

7.2 Discuss the impact

According to the data we have collected and preset, the preliminary calculation of the model can obtain the following 3 quantitative charts on selection and impact under time series analysis.

Under TAPGM, the government is in the dominant position. Therefore, we give priority to the government's choice.

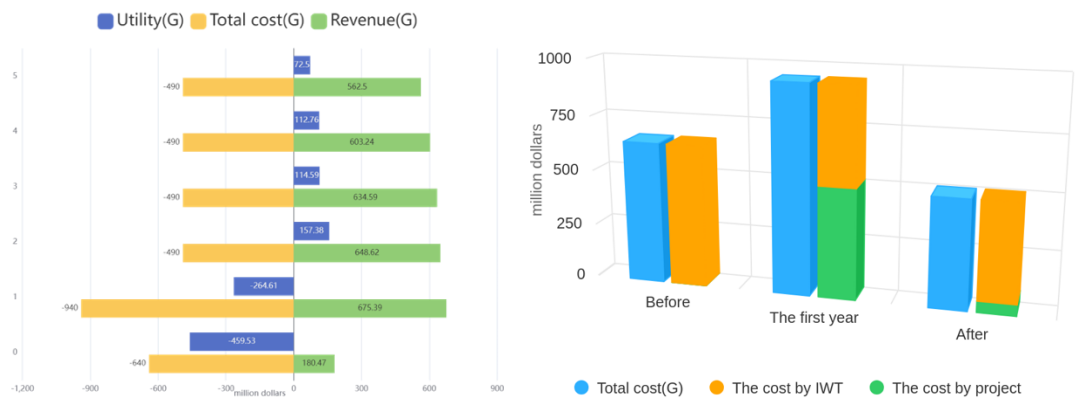

Figure 10: The utility structure of the government (Left), The cost structure of the government (Right)

According to the Figure 10, before the government implemented the project, due to the negative external effects caused by poaching, which led to a decrease in social welfare, the government's utility function was negative at this moment: $U_G = -460$.

After the implementation of the project, in the first year, the government needs to establish institutions and infrastructure. The total utility has increased compared with that in the first year, but it is still negative.

It can be seen in the Figure 10 that: Although it has exacerbated the fiscal deficit, the cost caused by IWT decreased from about 640 to 440. In the long run, once the asset investment in the first year is completed, it will only need to spend about 50million dollars each year to

maintain the regular operation of the project in the future.

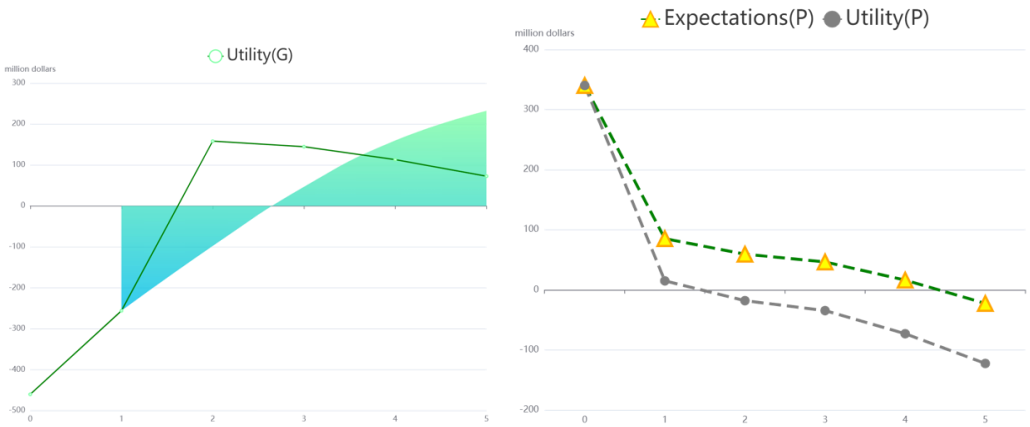

Figure 11: The accumulated benefits of government (Left),
The expectations and utility of poachers (Right)

In Figure 11, the annual utility from the second year to the fifth year is positive, and the accumulated benefits ($U_{G-5} = 488$)in the following years can gradually **make up** for the loss of utility ($U_{G-0} = -256$) in the previous years. The net utility of the government $U_{G-2.3.4} = 232$.Therefore, the implementation of this measure by the government is conducive to the **long-term development of national welfare**.

Next, see the choice of poachers according to figure 12. When the government's action is given (that is, the government decides to implement project), $E_P = 0$.

If poaching is still carried out, the expectation is positive in the first four years and negative from the fifth year ($E_{P-5} = -22.6$). Even if the expectation in the first four years is positive, it has also been showing a monotonous decreasing trend.

As poachers are rational economic men, we can reasonably expect that: Once there are employment opportunities provided economic profits in the society (higher than the expectations of poaching), so that the **opportunity cost** of poaching exceeds the utility obtained. Most poachers will gradually stop poaching with the implementation of the project. Until the fifth year, the regional illegal animal trade can be largely eliminated.

In the model, we also discuss the effectiveness of the **positive externalities** of combating IWT. Therefore, it can also be reasonably inferred that the implementing project can promote the improvement of international image, the development of other domestic industries, and the improvement of national cultural quality to a certain extent.

We also discussed the results of **cooperation with other countries in decreasing the demands** of illegal wildlife products. This can also draw a conclusion: the implementation of the project can combat IWT worldwide and call for more countries to take actions.

8 Task 5: How likely is the project to be successful?

In summary, the client should adopt our project. Based on the analysis results of Task 2 and Task 3, we know that the target client needs to rely on the help of developed countries and international organizations, such as the World Bank, to achieve the expected goals of the project. With the help of external forces, we need to consider more factors and the situation becomes

very complex. When using traditional BP neural networks for fitting and merging predictions, the weights and thresholds are easily trapped in local optima. Therefore, based on Task 2, we use an **Improved Genetic Algorithm (IGA)** to optimize the BP neural network and avoid its local convergence.

The specific content of our IGA is as follows:

1. We adopt real number encoding and use the reciprocal of mean square error as the fitness function:

$$F(x) = \frac{1}{MSE} \tag{18}$$

2. For the selection operation, we first calculate the fitness value of each individual, and then sort these individuals in order based on their fitness value. The first two individuals with the highest fitness value are directly inherited to the next generation, and the remaining individuals are evenly divided into three levels based on their fitness: first, second, and third. The first level individual replicates 2 copies to pass on to the next generation, the second level individual replicates 1 copy to pass on to the next generation, and the third level individual discards it and does not use it. This method **improves the average fitness of the population** and maintains its diversity.

3. For the probability of crossover and mutation, traditional genetic algorithms maintain the same probability of crossover and mutation occurring each iteration, while adaptive genetic algorithms can adaptively adjust the probability of crossover and mutation:

$$P_c = \begin{cases} \dfrac{k_1(f_{best} - f)}{f_{best} - f_{avg}}, & f' \geq f_{avg} \\ k_2, & f' < f_{avg} \end{cases} \tag{19}$$

$$P_m = \begin{cases} \dfrac{k_3(f_{best} - f')}{f_{best} - f_{ave}}, & f \geq f_{avg} \\ k_4, & f < f_{avg} \end{cases} \tag{20}$$

In the above two equations, f_{max} is the maximum fitness value in the population, f_{avg} is the average fitness value in the population, f' is the larger fitness value for crossover, and f is the fitness value for individual variation.

And our genetic algorithm introduces the average fitness value $E(X)$ of the population and the degree of dispersion $D(X)$ of the fitness value to define **the coefficient of difference $\boldsymbol{\lambda}$**, adopt λ to optimize the formulas for crossover probability and mutation probability:

$$P_c = \begin{cases} \dfrac{k_1}{1 + e^{\frac{k_2}{\lambda}}} \times \dfrac{f_{best} - f'}{f_{best} - f_{avg}} + k_3, & f' \geq f_{avg} \\ k_4, & f' < f_{avg} \end{cases} \tag{21}$$

$$P_m = \begin{cases} \dfrac{k_5}{1+e^{\frac{k_6}{\lambda}}} \times \dfrac{f_{best}-f}{f_{best}-f_{avg}} + k_7, & f \geq f_{avg} \\ k_8, & f < f_{avg} \end{cases} \tag{22}$$

Among them:

$$\lambda = \frac{E(X)+1}{\sqrt{D(X)}} \tag{23}$$

$$E(X) = f_{avg} = \frac{f_1 + f_2 + \cdots + f_M}{M} \tag{24}$$

$$D(X) = \frac{f_1^2 + f_2^2 + \cdots + f_M^2}{M} - f_{avg}^2 \tag{25}$$

In the above equations, f_i represents the value of population fitness. As the population evolves, the average fitness increases and the degree of difference between populations decreases, resulting in a coefficient of difference λ continuously increasing, it can be concluded that $\frac{k_1}{1+e^{\frac{k_2}{\lambda}}} \in (0, \frac{k_1}{2})$, so $P_c \in (k_3, \frac{k_1}{2} + k_3)$, similarl y, it can be inferred that $P_m \in (k_7, \frac{k_5}{2} + k_7)$.

The above improvements to the genetic algorithm prevent the population from forming local convergence and falling into local optima during initial evolution. The process of optimizing the BP neural network through the improved genetic algorithm is shown in the following figure:

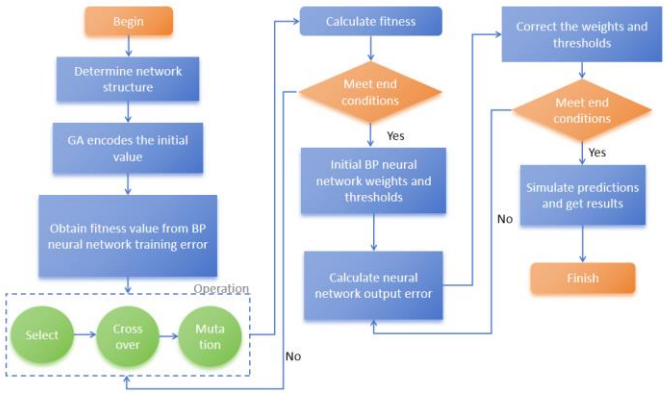

Figure 12: The flowchart of IGA optimized BP neural network

Below is an evaluation of the likelihood of the project achieving its expected goals.

Firstly, set our expected goals. For our clients, four core factors were selected for evaluation, with the number of poaching incidents G_N (Number of Poaching Incidents) reduced by 60%, illegal wildlife trading volume G_I (IWT Volume) reduced by 50%, environmental species evenness G_S (Specifications Evenness) increased by 5%, community engagement G_C (Community Engagement) increased by 10%.

Next, we collected some of the required data from the **World Bank official website** and **related databases**, and preprocessed the data, including filling in missing values and cleaning outliers. Finally, we obtained a complete set of four core factor data for 20 years.

Although neural networks are often regarded as “black box models”, in this problem, I use

the idea of sensitivity analysis to train the data for each year, so that we can understand the impact of different inputs on the model's prediction results. For our existing 20 years of data, we set the inputs of the neural network as: funding investment, project self-designed degree, and technological level for each year. Four core factors were used as outputs and fitted into the BP neural network to predict the changes of the four core factors in the future for clients seeking and not seeking additional assistance.

Taking capital investment as an example, the partial prediction results are plotted as follows:

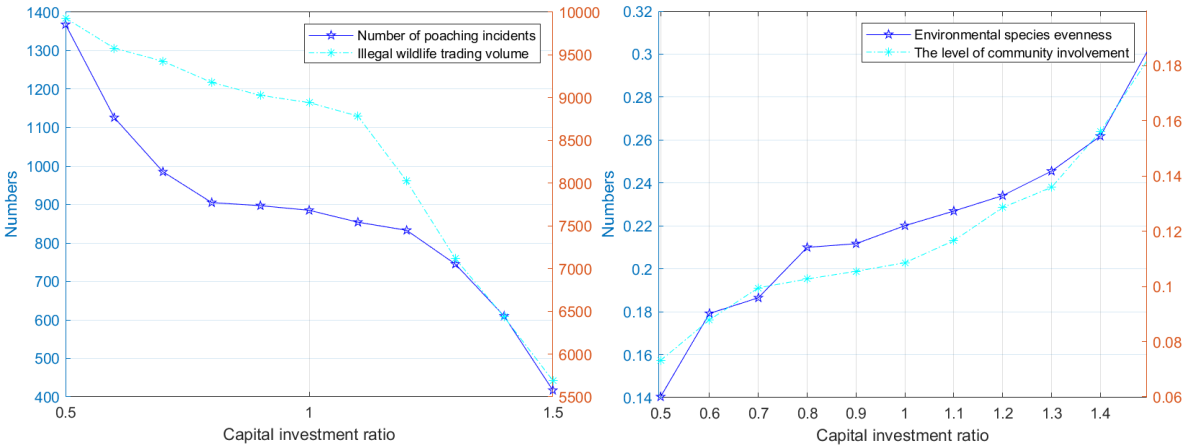

Figure 13: The relationship between the four core factors and capital investment

The above two graphs show the changes in the four core factors when the capital investment changes.

By comparing all the predicted results of Matlab with our expected goals, we can ultimately draw the following conclusion:

When the other two indicators remain unchanged:

①When the investment in funds increases by **20%**, we expect an **87.9%** chance of achieving the expected target; When the investment in funds increases by more than **37.1%**, we expect a **99%** chance of achieving the target.

②When the project design autonomy decreases by **5%**, we expect a **73.6%** chance of achieving the goal; When the project design autonomy decreases by more than **24%**, we expect a **99%** chance of achieving the goal.

③When the technological level improves by **5%**, we expect a **68.4%** chance of achieving the target; When the technological level improves by **16%**, we expect a **99%** chance of achieving the target.

Based on the above conclusion, further analysis shows that the indicator of technological level has a significant impact on the likelihood of achieving goals, which means that the likelihood of achieving goals is more sensitive to technological level. Therefore, if the target customers are unable to obtain better technological resources from advanced developed countries, it will form a certain degree of obstacle to the project's ability to achieve its goals; If the target customer can smoothly obtain good technical resources from advanced developed countries, it will help the project's ability to achieve its goals.

9 Sensitivity Analysis

To evaluate the sensitivity of various parameters in the model, we conducted a series of simulation experiments. For our intervention model, we adjusted the coefficients of a single intervention variable while fixing the coefficients of other intervention variables, and observed the changes in the model output. The obtained results show a linear relationship with a relatively flat slope, indicating that our intervention model has strong robustness, that is, small changes in parameters will not lead to significant differences in output results.

In the optimization of the BP neural network for the improved genetic algorithm (IGA), we adjusted the initial population size in IGA. The experimental results show that changes in population size have little impact on the accuracy of BP neural network prediction, that is, the prediction error is maintained at a low level on different test sets.

These analysis results all point to a common conclusion: **our model exhibits high stability**, that is, the output of the model remains relatively stable under changes in key parameters, further verifying the reliability and practicality of the model.

10 Model Evaluation and Further Discussion

10.1 Strengths

- The prediction accuracy of the BP neural network optimized by IGA is very high, and the prediction effect is very stable
- We have placed IWT in Africa within a **global market system**, considering the interactions with other economic entities and discussing how the impact of projects changes under such interactions.
- We can use game theory methods not only to evaluate the impact of the project, but also to **further persuade** the government to choose our project, presenting the cost and benefit changes of the project more clearly.

10.2 Weaknesses

- We only considered the impact of the market system and treated the other positive externalities brought by the project as parameters, lacking further discussion or consideration of the synergistic opportunities for other unexpected actors.
- We haven't analyzed some monopolistic behaviors that may exist in the market.

10.3 Further Discussion

- We can further analyze the national conditions and animal protection status of Southeast Asian countries and China in detail to improve our model and make it transplantable.
- We can also consider other external variables, such as the ecological environment, in the prediction model.
- We can further analyze whether there will be a change in equilibrium point if there is a monopoly or in a more complex system.

11 Conclusion

Our project presents a **robust and adaptive** approach to combating IWT, leveraging advanced technologies and regional cooperation. Through meticulous analysis and **innovative strategies**, we demonstrate a significant potential to reduce poaching incidents and foster a sustainable ecosystem, ensuring a harmonious balance between wildlife conservation and societal development.

12 Memo

From: Team 2416809, ICM 2024
To:Governments of southeast Africa
Date: February 5, 2024
Subject:Five year project --- using advanced technology to combat illegal wildlife trade
Dear leaders,

Illegal Wildlife Trade (IWT) is a global problem, which not only poses a serious threat to ecosystem diversity, but also endangers national economic development, especially in southeast Africa. We are well aware of the challenges your countries face in this regard. Therefore, we have put forward a five-year project proposal to make the action against illegal wildlife trade more effectively through technological innovation and regional cooperation. Now let me briefly introduce our project.

1. Deep learning and DNA tracking: Combine deep learning algorithm and DNA technology, analyze and predict illegal trading behavior and trace the source of wild animals to effectively combat criminal gangs;

2. Drone inspection and three-level management system: Equipping personnel with drones and establishing a three-level supervision system, not only does it improve regulatory efficiency, but it also reduces the possibility of corruption in grassroots law enforcement;

3. Digital sharing platform and public participation mechanism: Establish a digital sharing platform to allow the public to participate in wildlife conservation, while incentivizing the public to report illegal behavior through reward mechanisms;

The reason for choosing Southeast African countries as project implementation sites:

1. This area is a concentrated reflection of global biodiversity, with abundant wildlife resources, but at the same time, it faces serious threats from illegal wildlife trade;

2. This area has vast grasslands, which provides an ideal implementation environment for the use of UAV remote sensing monitoring and DNA tracking technology;

3. The goal of this project is closely dependent on your countries' mission to protect wildlife and maintain ecological balance.

Our project can bring you:

1. By improving technological means, it enhances monitoring of illegal poaching and trade activities;

2. Establishing a shared cooperation platform and reward mechanism can enhance public awareness of protection and increase community participation;

3. By strengthening intergovernmental cooperation, we can jointly combat illegal trade and enhance international cooperation and exchanges

We believe that through our cooperation, we can bring substantial progress to your countries and the entire southeastern region of Africa in protecting wildlife and combating illegal trade. We look forward to further discussing this project proposal with your countries and working together to contribute to the maintenance of biodiversity on Earth.

Regards,

Team 2416809

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Report on Use of AI

1. OpenAI ChatGPT (Nov 5, 2023 version, ChatGPT-4)
Query1: <Where can I get the data of illegal wildlife trafficking?>
Output: <To obtain data on illegal wildlife trade, you can explore multiple sources, including:

1. International Organizations and Databases
- **International Union for Conservation of Nature (IUCN)**: Offers data on species threats and conservation status, occasionally touching on illegal trade aspects.
- **Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES)**: Manages and monitors international trade in endangered species, regularly publishing reports and data.
- **World Wildlife Fund (WWF)**: Regularly publishes reports on wildlife conservation and illegal trade, providing relevant data and research.
- **INTERPOL**: As an international police organization, it publishes reports and data on transnational wildlife crime.
- **Global Environment Facility (GEF)**: Supports projects and activities aimed at protecting the global environment, including combating illegal wildlife trade.

2. Governments and Law Enforcement Agencies
- **National environmental protection or natural resources departments**: Responsible for wildlife protection and combating illegal trade, they may release relevant data and reports.
- **Customs and border protection agencies**: Customs data in some countries can reveal seizures of illegal wildlife products.

3. Research Institutions and Academic Journals
- **Universities and research institutions**: Scholars engaged in wildlife trade research might publish their findings in academic journals.
- **Professional journals**: Journals like "Biodiversity and Conservation" and "The Wildlife Society" often publish research articles on illegal wildlife trade.

4. Non-Governmental Organizations (NGOs)
- **Wildlife conservation organizations**: Organizations like the Wildlife Conservation Society (WCS) and the Animal Welfare Institute often conduct investigations and publish related reports and data.

5. Social Media and News Reports
- **News media**: News reports covering illegal wildlife trade cases and investigation results can provide valuable information.
- **Social media platforms**: Sometimes, relevant organizations or individuals share real-time information or cases of illegal trade.

When sourcing data, ensure the reliability and credibility of the information and its source to ensure accuracy and legality. Given the sensitivity and complexity of illegal wildlife trade, some data might be difficult to access or confidential.>